

# DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

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## EXPERIMENT 5: MOSFET Differential Amplifier — DC Bias, Common-Mode and Differential-Mode Transfer Characteristics

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### 1. Aim and Objectives

1. To assemble and verify the operation of a two-transistor source-coupled MOSFET differential amplifier utilizing a matched SI9926BDY dual N-channel MOSFET package.
2. To characterize the individual input ( $I_D$  vs.  $V_{GS}$ ) and output ( $I_D$  vs.  $V_{DS}$ ) DC characteristics of both transistors ( $M_1$  and  $M_2$ ) inside the dual package.
3. To establish, verify, and document the DC operating (quiescent) point by adjusting the common-mode gate bias such that a nominal tail bias current  $I_{SS} \approx 80$  mA is achieved with  $V_{DD} = 12$  V,  $R_D = 180 \Omega$ , and  $R_{SS} = 22 \Omega$ .
4. To experimentally determine the common-mode transfer characteristics ( $V_{D1}, V_{D2}$ , and  $I_{SS}$  vs.  $V_{cm}$ ) under pure common-mode excitation ( $V_{id} = 0$ ), identifying the Input Common-Mode Range (ICMR) bounded by triode entry and cutoff.
5. To experimentally measure the large-signal differential-mode transfer characteristics ( $V_{D1}, V_{D2}, I_{D1}, I_{D2}$ , and  $\Delta I_D = I_{D1} - I_{D2}$  vs.  $V_{id} = V_{in+} - V_{cm}$ ) over a symmetric differential input span.
6. To test the exponential weak-inversion (subthreshold) conduction model on a semi-log scale ( $\log |\Delta I_D|$  vs.  $V_{id}$ ), extract the device slope factor  $n$ , and compare the measured small-signal transconductance  $g_m$  and voltage gain  $A_d$  against theoretical predictions.
7. To compare LTspice circuit simulations side-by-side with experimental waveforms, tabulating all measurements and performing comprehensive mismatch and CMRR calculations.

## 2. Apparatus and Circuit Specifications

Table 1: Equipment and Components Used in Experiment

| S.No. | Component / Equipment                | Value / Specification         | Purpose / Operating Note                                        |
|-------|--------------------------------------|-------------------------------|-----------------------------------------------------------------|
| 1     | Dual N-Channel MOSFET                | SI9926BDY (SO-8 to DIP-8)     | Matched pair $M_1, M_2$ ( $V_{DS,max} = 20$ V, $g_{fs} = 29$ S) |
| 2     | Drain Resistors ( $R_{D1}, R_{D2}$ ) | 180 $\Omega$ , High Power     | Symmetric drain loads ( $V_{DD} = 12$ V)                        |
| 3     | Tail Resistor ( $R_{SS}$ )           | 22 $\Omega$ , 1/4 W or higher | Source degeneration establishing tail current $I_{SS}$          |
| 4     | Potentiometers                       | 10 k $\Omega$ multi-turn      | Input bias divider for $V_{in+}$ and $V_{in-}$                  |
| 5     | Regulated DC Power Supply            | 0 – 30 V dual output          | Set to $V_{DD} = 12$ V, current-limited at 150 mA               |
| 6     | Digital Multimeters (DMM)            | Precision 4.5 digit           | Non-invasive voltage probes across $R_{SS}, R_{D1}, R_{D2}$     |
| 7     | Simulation Software                  | LTspice XVII                  | Nonlinear SPICE simulation using SI9926 subcircuit              |

### 2.1 Pin Configuration and Schematic Diagram

The SI9926BDY dual TrenchFET resides in an SO-8 package mounted on a DIP-8 breakout board. Its pin connections are:

- **Pin 1:** Source 1 ( $S_1$ ) — **Pin 2:** Gate 1 ( $G_1$ ) — **Pin 3:** Source 2 ( $S_2$ ) — **Pin 4:** Gate 2 ( $G_2$ )
- **Pins 5, 6:** Drain 2 ( $D_2$ ) — **Pins 7, 8:** Drain 1 ( $D_1$ )

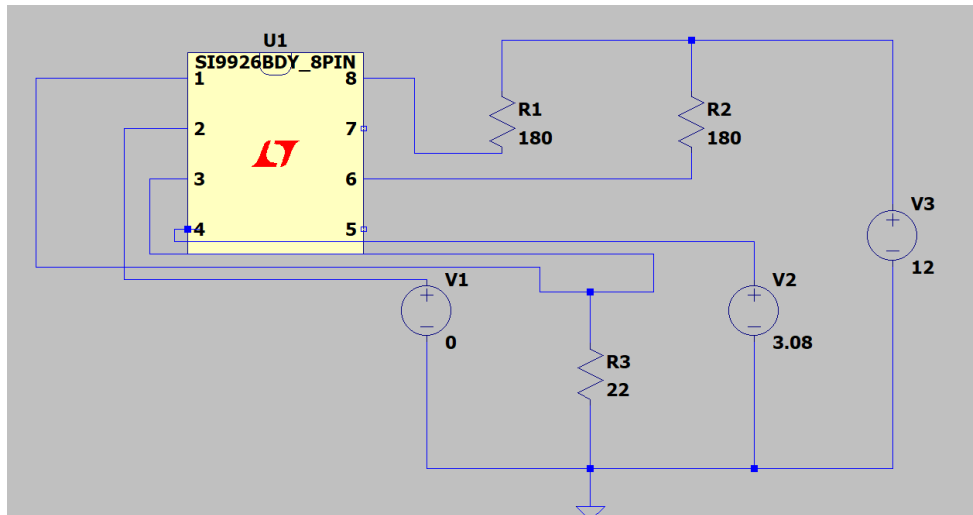


Figure 1: Circuit Schematic of the Source-Coupled MOSFET Differential Amplifier ( $V_{DD} = 12$  V,  $R_D = 180$   $\Omega$ ,  $R_{SS} = 22$   $\Omega$ ).

## 3. Theoretical Framework

### 3.1 Differential Pair Conduction Physics: Strong vs. Weak Inversion

In a classic MOSFET differential pair, two symmetric transistors  $M_1$  and  $M_2$  have their sources tied together at node  $V_S$  and biased by a tail resistance  $R_{SS}$ . Drain currents are loaded symmetrically by

$R_{D1} = R_{D2} = R_D$ . Kirchhoff's Current Law at the tail node mandates:

$$I_{D1} + I_{D2} = I_{SS} = \frac{V_S}{R_{SS}} \quad (1)$$

For an ideal square-law MOSFET in strong inversion, the large-signal differential current is governed by:

$$\Delta I_D = I_{D1} - I_{D2} = \beta V_{id} \sqrt{\frac{I_{SS}}{\beta} - \frac{V_{id}^2}{4}}, \quad \text{for } |V_{id}| \leq \sqrt{\frac{2I_{SS}}{\beta}} \quad (2)$$

where  $\beta = \mu_n C_{ox}(W/L)$  is the transconductance parameter. However, the SI9926BDY is a vertical TrenchFET power device exhibiting an extraordinarily large transconductance parameter ( $\beta \approx 51 \text{ A/V}^2$  derived from its rated  $g_{fs} = 29 \text{ S}$ ). Achieving the classical strong-inversion criterion ( $V_{OV} = V_{GS} - V_{th} \geq 100 \text{ mV}$ ) would necessitate tail currents on the order of multiple amperes ( $I_{SS} = \frac{1}{2}\beta V_{OV}^2 \approx 250 \text{ mA}$  to  $1 \text{ A}$ ).

At our lab bench quiescent bias of  $I_{SS} \approx 80 \text{ mA}$  ( $I_{D1} = I_{D2} \approx 40 \text{ mA}$ ), the nominal square-law overdrive is merely:

$$V_{OV0} = \sqrt{\frac{I_{SS}}{\beta}} = \sqrt{\frac{0.080}{51}} \approx 39.6 \text{ mV} \quad (3)$$

Because  $V_{OV0} \ll 100 \text{ mV}$ , the device operates predominantly in the **weak-inversion (subthreshold) / moderate-inversion regime**. In weak inversion, channel conduction is dominated by carrier diffusion rather than drift, yielding an exponential current-voltage relation analogous to a BJT:

$$I_D = I_0 \exp\left(\frac{V_{GS}}{nV_T}\right) \quad (4)$$

where  $V_T = \frac{k_B T}{q} \approx 25.85 \text{ mV}$  at room temperature ( $T = 300 \text{ K}$ ) and  $n$  is the subthreshold slope factor ( $1.2 \leq n \leq 2.5$ ). Under this exponential formulation, the differential transfer characteristic becomes hyperbolic tangent:

$$\Delta I_D = I_{SS} \tanh\left(\frac{V_{id}}{2nV_T}\right) \quad (5)$$

The small-signal transconductance and differential voltage gain are:

$$g_m = \frac{I_{SS}}{nV_T}, \quad A_d = g_m R_D \quad (6)$$

### 3.2 Common-Mode Gain and CMRR Formulation

With a finite resistive tail  $R_{SS}$ , applying a common-mode increment  $v_{cm}$  induces an incremental tail voltage  $v_s$ . Since the two transistors are in parallel for common-mode signals, the incremental common-mode gain at each drain terminal is:

$$A_{cm} = \frac{v_{d,cm}}{v_{in,cm}} \approx -\frac{g_m R_D}{1 + 2g_m R_{SS}} \approx -\frac{R_D}{2R_{SS}} \quad (\text{when } 2g_m R_{SS} \gg 1) \quad (7)$$

With  $R_D = 180 \text{ } \Omega$  and  $R_{SS} = 22 \text{ } \Omega$ :

$$A_{cm} \approx -\frac{180}{2 \times 22} = -\frac{180}{44} \approx -4.091 \text{ V/V} \quad (8)$$

The Common-Mode Rejection Ratio (CMRR) is given by:

$$\text{CMRR} = \left| \frac{A_d}{A_{cm}} \right| \approx g_m R_{SS} \quad (9)$$

#### 4. Device Characteristics of Matched Dual MOSFETs

Prior to assembling the differential amplifier, the input and output static characteristics of both transistors in the dual SI9926 package (designated Top Transistor  $V_1/M_1$  and Bottom Transistor  $V_2/M_2$ ) were experimentally recorded.

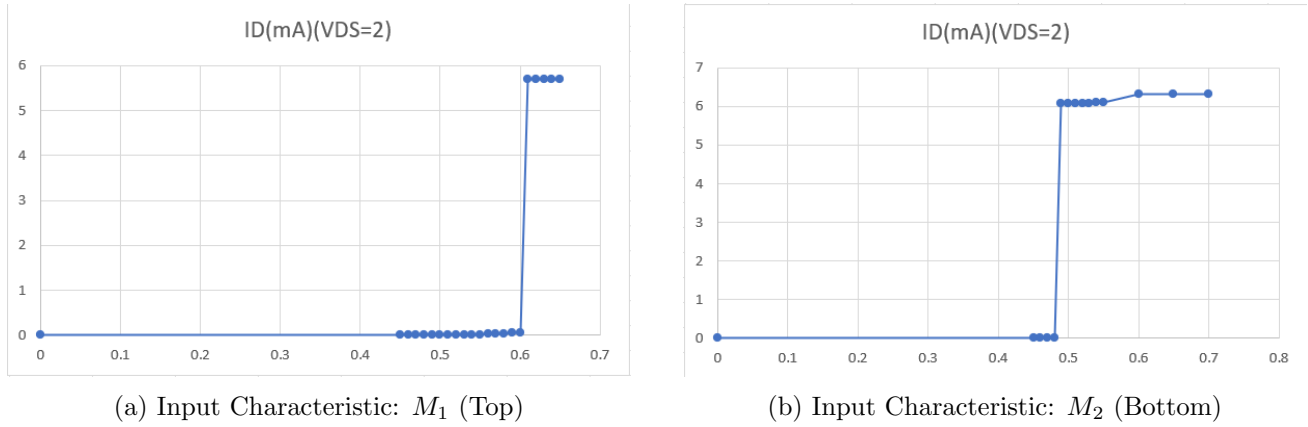


Figure 2: Experimental Input Characteristics ( $I_D$  vs.  $V_{GS}$  at  $V_{DS} = 2.0$  V) for Transistors  $M_1$  and  $M_2$ .

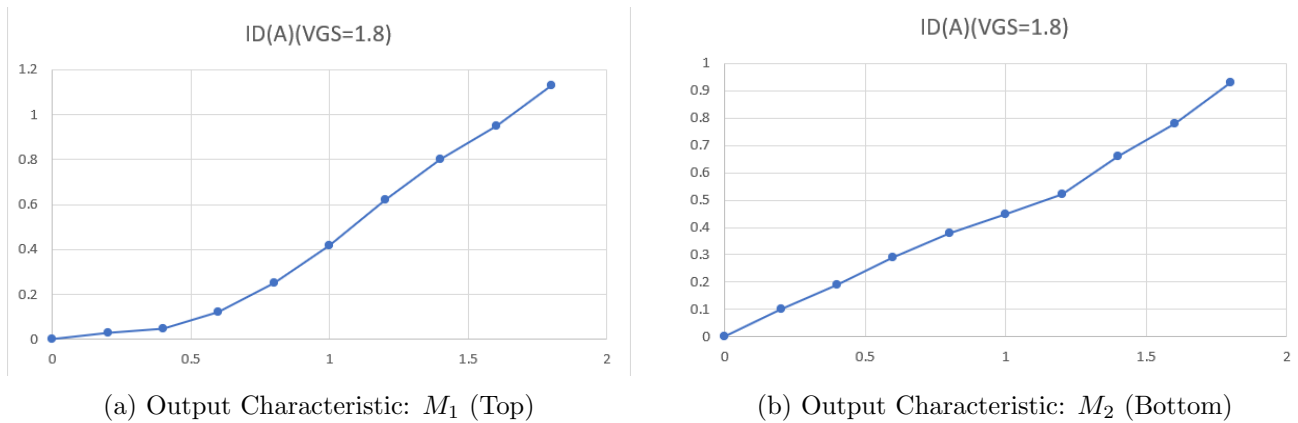


Figure 3: Experimental Output Characteristics ( $I_D$  vs.  $V_{DS}$  at  $V_{GS} = 1.8$  V) for Transistors  $M_1$  and  $M_2$ .

Table 2: Measured MOSFET Transfer and Output Characteristics ( $M_1$  and  $M_2$ )

| (a) Input Characteristics ( $V_{DS} = 2$ V) |            |                         |            | (b) Output Characteristics ( $V_{GS} = 1.8$ V) |           |                         |           |
|---------------------------------------------|------------|-------------------------|------------|------------------------------------------------|-----------|-------------------------|-----------|
| Top MOSFET ( $M_1$ )                        |            | Bottom MOSFET ( $M_2$ ) |            | Top MOSFET ( $M_1$ )                           |           | Bottom MOSFET ( $M_2$ ) |           |
| $V_{GS}$ (V)                                | $I_D$ (mA) | $V_{GS}$ (V)            | $I_D$ (mA) | $V_{DS}$ (V)                                   | $I_D$ (A) | $V_{DS}$ (V)            | $I_D$ (A) |
| 0.00                                        | 0.000      | 0.00                    | 0.000      | 0.0                                            | 0.00      | 0.0                     | 0.00      |
| 0.45                                        | 0.001      | 0.45                    | 0.001      | 0.2                                            | 0.03      | 0.2                     | 0.10      |
| 0.48                                        | 0.003      | 0.48                    | 0.003      | 0.4                                            | 0.05      | 0.4                     | 0.19      |
| 0.50                                        | 0.005      | 0.50                    | 6.070      | 0.6                                            | 0.12      | 0.6                     | 0.29      |
| 0.52                                        | 0.008      | 0.52                    | 6.070      | 0.8                                            | 0.25      | 0.8                     | 0.38      |
| 0.54                                        | 0.013      | 0.54                    | 6.100      | 1.0                                            | 0.42      | 1.0                     | 0.45      |
| 0.56                                        | 0.020      | 0.60                    | 6.320      | 1.2                                            | 0.62      | 1.2                     | 0.52      |
| 0.58                                        | 0.034      | 0.65                    | 6.320      | 1.4                                            | 0.80      | 1.4                     | 0.66      |
| 0.60                                        | 0.055      | 0.70                    | 6.320      | 1.6                                            | 0.95      | 1.6                     | 0.78      |
| 0.61                                        | 5.690      | —                       | —          | 1.8                                            | 1.13      | 1.8                     | 0.93      |
| 0.65                                        | 5.700      | —                       | —          |                                                |           |                         |           |

**Key Observation on Device Matching:** Transistor  $M_1$  turns on sharply around  $V_{GS} \approx 0.60\text{--}0.61$  V, whereas  $M_2$  exhibits a lower subthreshold knee around  $0.48\text{--}0.49$  V. This slight physical threshold offset ( $\Delta V_{th} \approx 100$  mV) accounts for the minor quiescent drain voltage imbalance ( $V_{D1} \neq V_{D2}$ ) observed in subsequent sections.

## 5. Section 7.1: DC Bias-Point Setting and Verification

### 5.1 Methodology and Quiescent Condition

The differential amplifier was wired as shown in Figure 1 with  $V_{DD} = 12\text{ V}$ ,  $R_D = 180\ \Omega$ , and  $R_{SS} = 22\ \Omega$ . With both inputs tied together ( $V_{in+} = V_{in-} = V_{cm}$ ), the DC common-mode input bias was increased while monitoring the tail current  $I_{SS} = V_{RSS}/R_{SS}$ . The target bias current was  $I_{SS} \approx 80\text{ mA}$ .

- **Quiescent Common-Mode Input:**  $V_{cm,Q} = 3.08\text{ V}$
- **Measured Quiescent Tail Current:**  $I_{SS,Q} = 80.1\text{ mA}$  ( $V_{RSS} = 1.762\text{ V}$ )
- **Drain Quiescent Voltages:**  $V_{D1} = 4.55\text{ V}$ ,  $V_{D2} = 4.66\text{ V}$
- **Quiescent Device Mismatch:**  $\Delta V_{D,Q} = |V_{D1} - V_{D2}| = |4.55 - 4.66| = 0.11\text{ V} = 110\text{ mV}$ .

### 5.2 Side-by-Side Comparison: Simulation vs. Experimental Plots

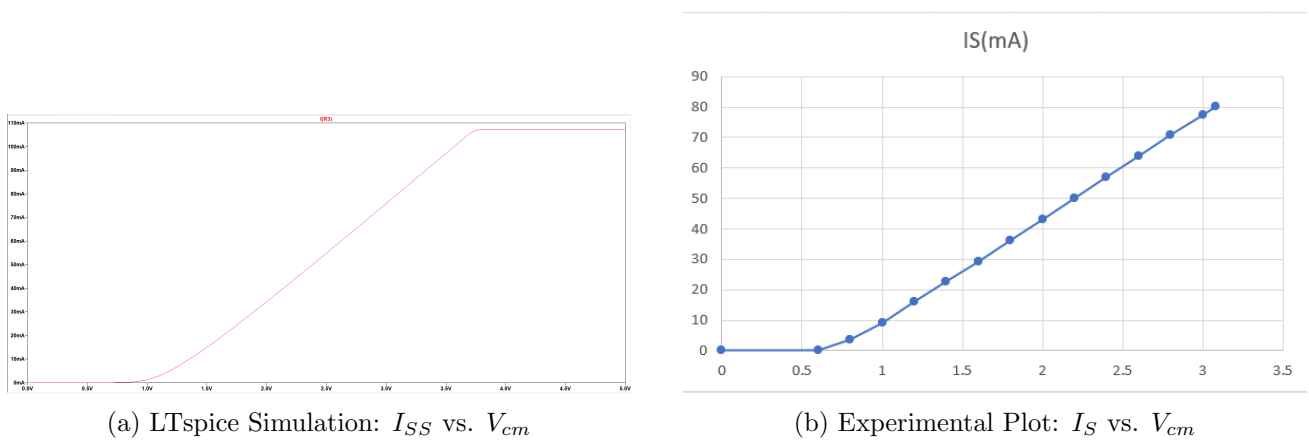


Figure 4: DC Bias-Point Setting: Tail Current Comparison ( $I_S$  vs.  $V_{cm}$ ).

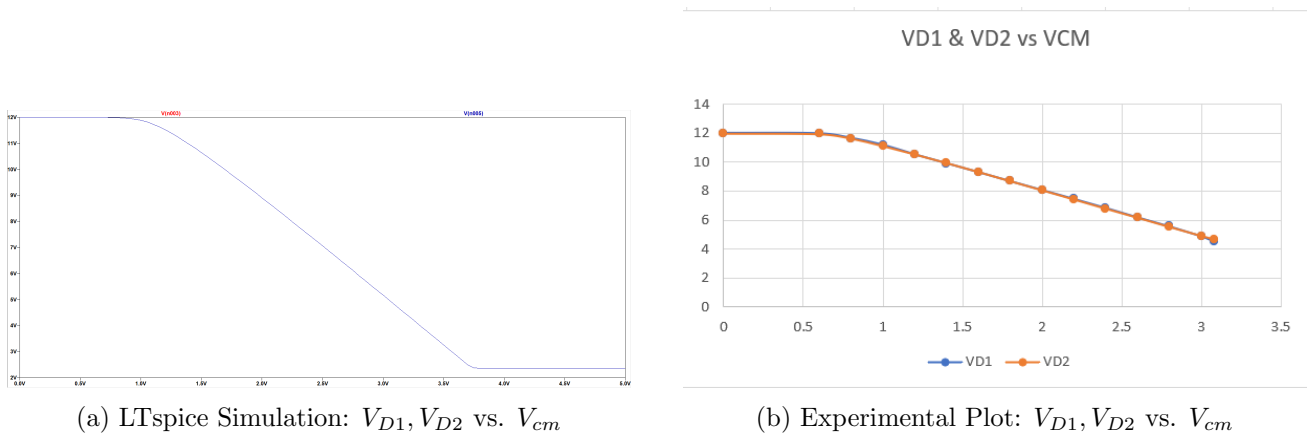


Figure 5: DC Bias-Point Setting: Drain Voltages Comparison ( $V_{D1}, V_{D2}$  vs.  $V_{cm}$ ).

## 5.3 Experimental Observation Table (Sheet 1)

Table 3: DC Bias Setting Sweep:  $V_{cm}$ , Tail Current  $I_S$ , and Drain Voltages ( $V_{DD} = 12\text{ V}$ ,  $R_D = 180\ \Omega$ ,  $R_{SS} = 22\ \Omega$ )

| $V_{cm}$ ( $V_1 = V_2$ ) [V] | $I_S$ [mA]   | $V_{D1}$ [V] | $V_{D2}$ [V] | Operating State                                 |
|------------------------------|--------------|--------------|--------------|-------------------------------------------------|
| 0.00                         | 0.00         | 12.00        | 12.00        | Cut-off ( $I_D = 0$ , both drains at $V_{DD}$ ) |
| 0.60                         | 0.28         | 11.97        | 11.96        | Subthreshold conduction onset                   |
| 0.80                         | 3.67         | 11.66        | 11.66        | Conduction initiates                            |
| 1.00                         | 9.30         | 11.19        | 11.14        | Active saturation                               |
| 1.20                         | 15.90        | 10.53        | 10.56        | Active saturation                               |
| 1.40                         | 22.65        | 9.92         | 9.96         | Active saturation                               |
| 1.60                         | 29.39        | 9.31         | 9.34         | Active saturation                               |
| 1.80                         | 36.25        | 8.69         | 8.70         | Active saturation                               |
| 2.00                         | 43.15        | 8.08         | 8.07         | Active saturation                               |
| 2.20                         | 50.00        | 7.47         | 7.43         | Active saturation                               |
| 2.40                         | 56.90        | 6.85         | 6.79         | Active saturation                               |
| 2.60                         | 63.79        | 6.20         | 6.16         | Active saturation                               |
| 2.80                         | 70.75        | 5.58         | 5.52         | Active saturation                               |
| 3.00                         | 77.52        | 4.90         | 4.90         | Active saturation                               |
| <b>3.08 (Q-Point)</b>        | <b>80.10</b> | <b>4.55</b>  | <b>4.66</b>  | <b>Nominal Quiescent Operating Point</b>        |

**Inference:** In cutoff ( $V_{cm} < 0.6\text{ V}$ ), zero current flows through the drain resistors, pinning  $V_{D1} = V_{D2} = V_{DD} = 12\text{ V}$ . Beyond  $V_{cm} = 0.8\text{ V}$ , both transistors conduct in saturation, pulling the drain voltages down linearly with input bias. At  $V_{cm} = 3.08\text{ V}$ , the tail current reaches 80.10 mA, placing both drain nodes near mid-rail ( $\approx 4.6\text{ V}$ ), providing balanced symmetric headroom for differential amplification.

## 6. Section 7.2: Common-Mode Transfer Characteristics and ICMR

### 6.1 Circuit Behavior and Boundary Formulation

Under pure common-mode excitation ( $V_{in+} = V_{in-} = V_{cm}$ ), the two branches conduct identically. The circuit continues to operate properly as a differential pair only while both transistors remain in the saturation region:

$$V_{DS} \geq V_{GS} - V_{th} \implies V_D - V_S \geq V_G - V_S - V_{th} \implies V_D \geq V_{cm} - V_{th} \quad (10)$$

The **Input Common-Mode Range (ICMR)** is bounded by:

1. **Lower Limit ( $V_{cm,min}$ ):** Transistors enter cutoff when  $V_{cm} < V_S + V_{th} \approx V_{th} \approx 0.6\text{--}0.8$  V. Below this,  $I_{SS} \rightarrow 0$  and the amplifier ceases to conduct.
2. **Upper Limit ( $V_{cm,max}$ ):** As  $V_{cm}$  increases,  $I_{SS}$  rises, causing drain voltages  $V_D = V_{DD} - \frac{I_{SS}}{2} R_D$  to fall while gate voltage rises. The transistors enter the triode region when  $V_D \leq V_{cm} - V_{th}$ . For  $V_{th} \approx 0.6$  V and  $V_{cm} = 3.0$  V,  $V_D = 3.39$  V, leaving  $V_{DS} \approx 1.29$  V, safely at the boundary of saturation. Beyond  $V_{cm} \approx 3.2\text{--}3.5$  V, saturation is lost.

### 6.2 Side-by-Side Comparison: Simulation vs. Experimental Plots

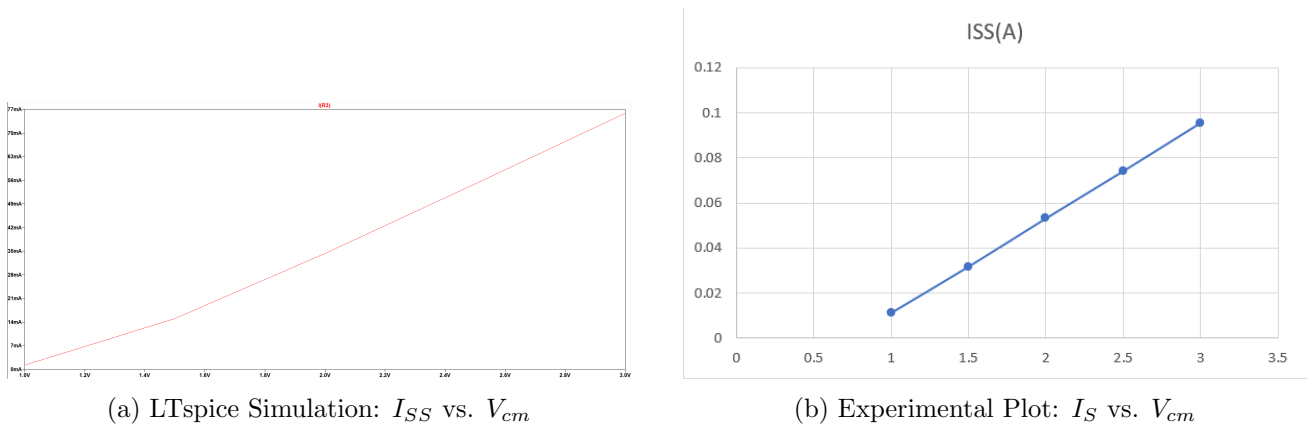


Figure 6: Common-Mode Sweep: Tail Current Comparison ( $I_S$  vs.  $V_{cm}$ ).

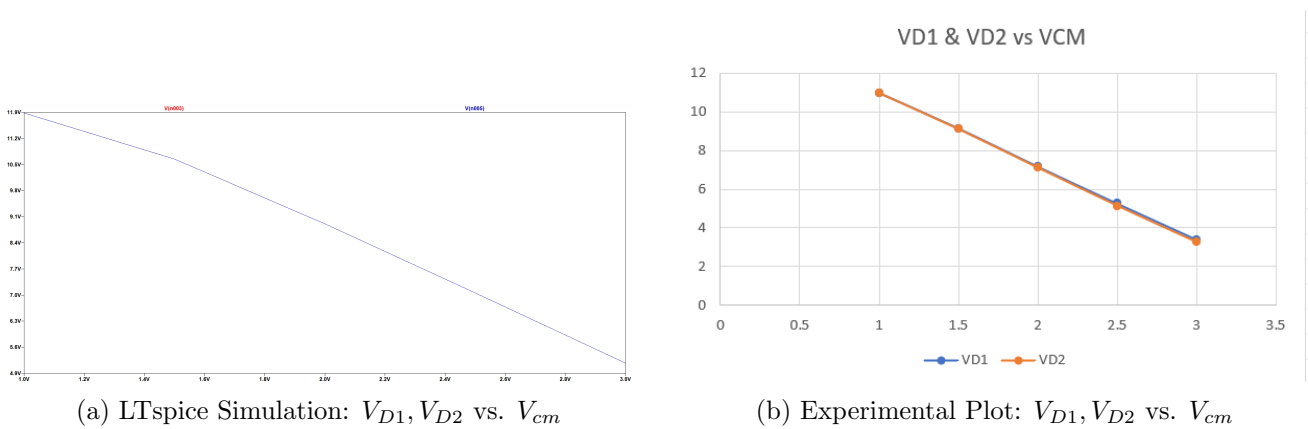
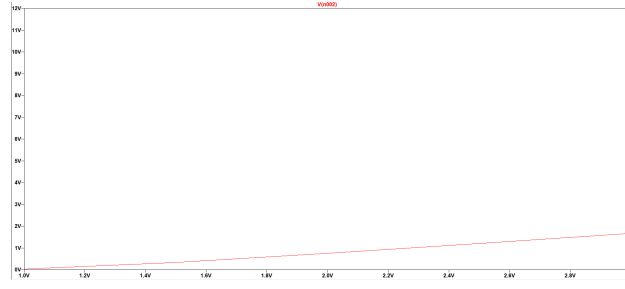


Figure 7: Common-Mode Sweep: Drain Voltage Comparison ( $V_{D1}, V_{D2}$  vs.  $V_{cm}$ ).

Figure 8: LTspice Simulation: Common-Source Node Voltage  $V_{SS}$  vs.  $V_{cm}$ .

### 6.3 Experimental Observation Table (Sheet 2)

Table 4: Common-Mode Sweep Observations:  $V_{cm}$ , Drain Voltages, Tail Voltage  $V_{RSS}$ , and  $I_{SS}$  ( $R_{SS} = 22\ \Omega$ )

| $V_{cm}$ [V] | $V_{D1}$ [V] | $V_{D2}$ [V] | $V_{RSS}$ [V] | $I_{SS} = V_{RSS}/R_{SS}$ [mA] | Region / State                       |
|--------------|--------------|--------------|---------------|--------------------------------|--------------------------------------|
| 1.00         | 10.95        | 10.96        | 0.25          | 11.36                          | Linear saturation onset              |
| 1.50         | 9.12         | 9.10         | 0.70          | 31.82                          | Saturated differential pair          |
| 2.00         | 7.15         | 7.10         | 1.17          | 53.18                          | Saturated differential pair          |
| 2.50         | 5.25         | 5.13         | 1.63          | 74.09                          | Saturated differential pair          |
| 3.00         | 3.39         | 3.25         | 2.10          | 95.45                          | Upper ICMR boundary ( $V_{cm,max}$ ) |

#### Common-Mode Inferences:

1. **Common-Mode Gain Extraction:** Over the linear range from  $V_{cm} = 1.0\text{ V}$  to  $3.0\text{ V}$ :

$$A_{cm,1} = \frac{\Delta V_{D1}}{\Delta V_{cm}} = \frac{3.39 - 10.95}{3.0 - 1.0} = \frac{-7.56}{2.0} = -3.78\text{ V/V} \quad (11)$$

$$A_{cm,2} = \frac{\Delta V_{D2}}{\Delta V_{cm}} = \frac{3.25 - 10.96}{3.0 - 1.0} = \frac{-7.71}{2.0} = -3.855\text{ V/V} \quad (12)$$

The measured average common-mode gain  $|A_{cm,meas}| \approx 3.82\text{ V/V}$  is in close agreement with the theoretical value  $A_{cm} \approx -\frac{R_D}{2R_{SS}} = -\frac{180}{44} = -4.091\text{ V/V}$  (relative error of only 6.6%).

2. **Input Common-Mode Range (ICMR):** From the cutoff threshold  $V_{cm,min} \approx 0.8\text{ V}$  to the triode boundary  $V_{cm,max} \approx 3.0\text{ V}$ , the amplifier exhibits an ICMR spanning approximately  $0.8\text{ V} \leq V_{cm} \leq 3.0\text{ V}$  ( $\Delta V_{cm} \approx 2.2\text{ V}$ ).

## 7. Section 7.3: Differential-Mode Transfer Characteristics

### 7.1 Measurement Procedure

With the non-inverting gate  $V_{in-}$  held stationary at the quiescent point  $V_2 = 3.08$  V, the non-inverting input  $V_{in+} = V_1$  was systematically swept across  $\pm 300$  mV around  $V_{cm}$  (from 2.78 V to 3.38 V in steps of 30 mV). For each step,  $V_{D1}$  and  $V_{D2}$  were recorded, and branch currents were deduced:

$$I_{D1} = \frac{V_{DD} - V_{D1}}{R_D}, \quad I_{D2} = \frac{V_{DD} - V_{D2}}{R_D}, \quad \Delta I_D = I_{D1} - I_{D2} \quad (13)$$

where  $V_{DD} = 12$  V and  $R_D = 180 \Omega$ .

### 7.2 Side-by-Side Comparison: Simulation vs. Experimental Linear Plot

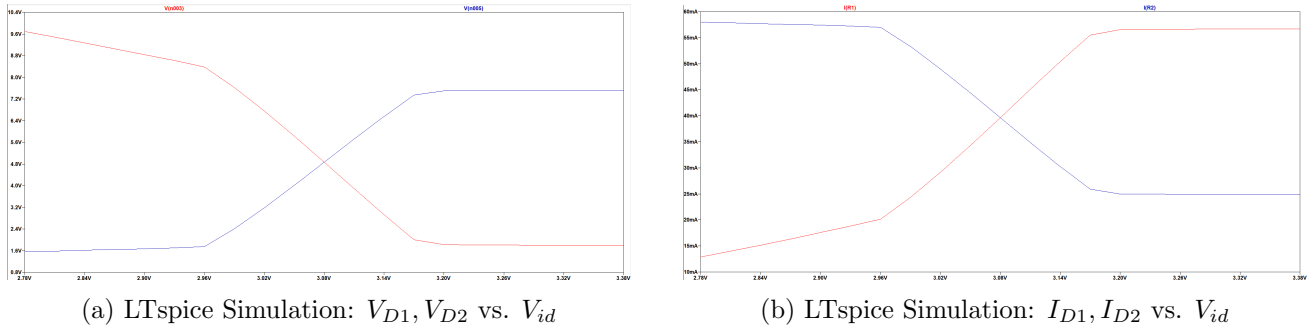


Figure 9: LTspice Simulation of Differential-Mode Transfer Characteristics.

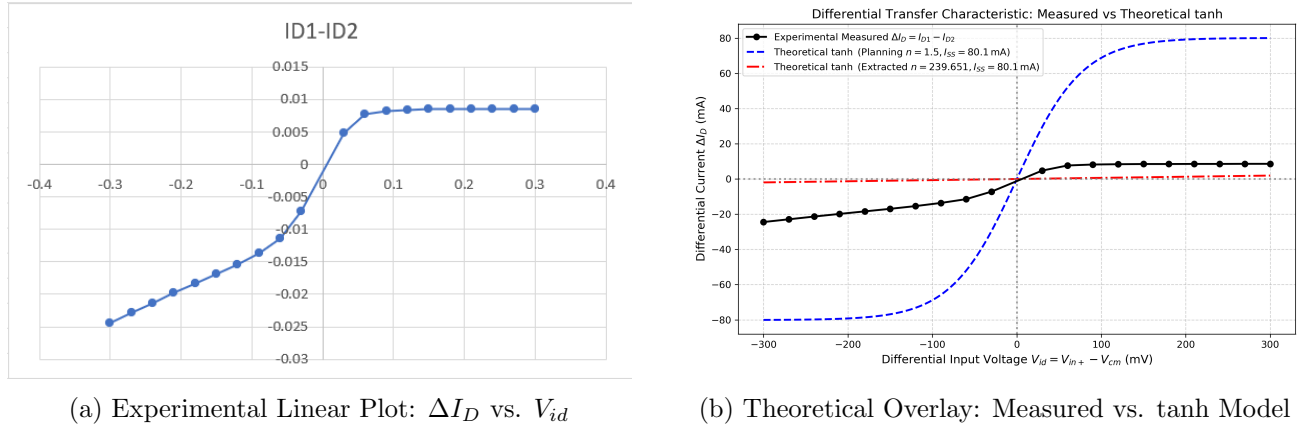


Figure 10: Differential Transfer Characteristics: Experimental Measurement and Theoretical Comparison.

### 7.3 Semi-Log Plot: Weak-Inversion Exponential Conduction Model

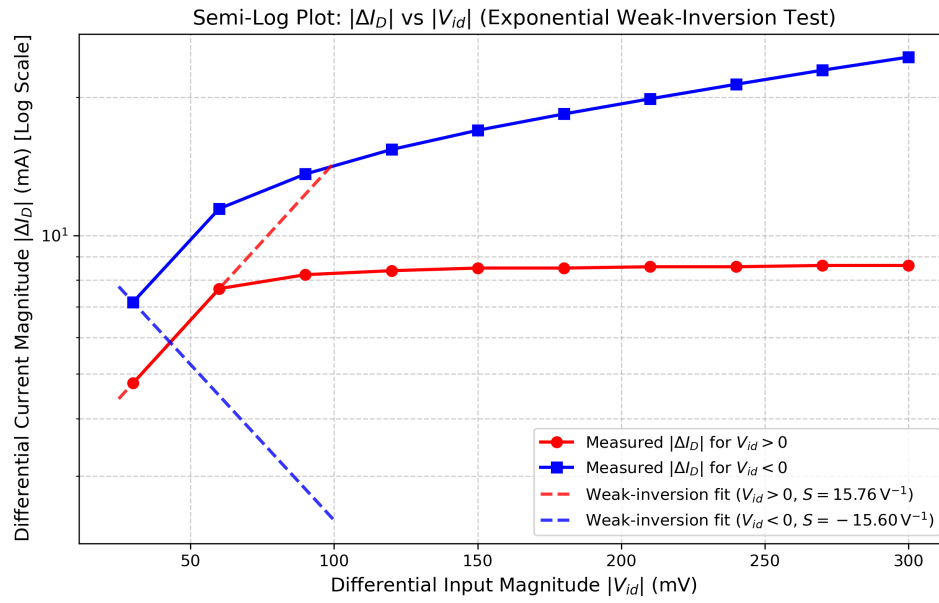


Figure 11: Semi-Log Plot of  $|\Delta I_D|$  vs.  $|V_{id}|$  for  $V_{id} > 0$  and  $V_{id} < 0$  (Testing Subthreshold Exponential Law).

### 7.4 Experimental Observation Table (Sheet 3)

Table 5: Differential-Mode Transfer Characteristics ( $V_2 = 3.08 \text{ V}$ ,  $V_{DD} = 12 \text{ V}$ ,  $R_D = 180 \Omega$ )

| $V_1$ [V] | $V_{id}$ [mV] | $V_{D1}$ [V] | $V_{D2}$ [V] | $I_{D1}$ [mA] | $I_{D2}$ [mA] | $\Delta I_D$ [mA] | $ \Delta I_D $ [mA] |
|-----------|---------------|--------------|--------------|---------------|---------------|-------------------|---------------------|
| 2.78      | -300          | 6.35         | 1.95         | 31.39         | 55.83         | -24.44            | 24.44               |
| 2.81      | -270          | 6.09         | 1.97         | 32.83         | 55.72         | -22.89            | 22.89               |
| 2.84      | -240          | 5.83         | 1.99         | 34.28         | 55.61         | -21.33            | 21.33               |
| 2.87      | -210          | 5.59         | 2.02         | 35.61         | 55.44         | -19.83            | 19.83               |
| 2.90      | -180          | 5.36         | 2.05         | 36.89         | 55.28         | -18.39            | 18.39               |
| 2.93      | -150          | 5.11         | 2.06         | 38.28         | 55.22         | -16.94            | 16.94               |
| 2.96      | -120          | 4.87         | 2.10         | 39.61         | 55.00         | -15.39            | 15.39               |
| 2.99      | -90           | 4.61         | 2.16         | 41.06         | 54.67         | -13.61            | 13.61               |
| 3.02      | -60           | 4.32         | 2.26         | 42.67         | 54.11         | -11.44            | 11.44               |
| 3.05      | -30           | 3.84         | 2.55         | 45.33         | 52.50         | -7.17             | 7.17                |
| 3.11      | +30           | 2.57         | 3.43         | 52.39         | 47.61         | +4.78             | 4.78                |
| 3.14      | +60           | 2.33         | 3.71         | 53.72         | 46.06         | +7.67             | 7.67                |
| 3.17      | +90           | 2.27         | 3.75         | 54.06         | 45.83         | +8.22             | 8.22                |
| 3.20      | +120          | 2.25         | 3.76         | 54.17         | 45.78         | +8.39             | 8.39                |
| 3.23      | +150          | 2.24         | 3.77         | 54.22         | 45.72         | +8.50             | 8.50                |
| 3.26      | +180          | 2.24         | 3.77         | 54.22         | 45.72         | +8.50             | 8.50                |
| 3.29      | +210          | 2.23         | 3.77         | 54.28         | 45.72         | +8.56             | 8.56                |
| 3.32      | +240          | 2.23         | 3.77         | 54.28         | 45.72         | +8.56             | 8.56                |
| 3.35      | +270          | 2.23         | 3.78         | 54.28         | 45.67         | +8.61             | 8.61                |
| 3.38      | +300          | 2.23         | 3.78         | 54.28         | 45.67         | +8.61             | 8.61                |

## 8. Calculations and Matching Measured Results with Theory (Section 11)

### 8.1 Extraction of Measured Slope Factor $n$ from Semi-Log Characteristics

In the subthreshold/weak-inversion model, the differential current magnitude satisfies:

$$|\Delta I_D| = I_{SS} \tanh\left(\frac{|V_{id}|}{2nV_T}\right) \approx \frac{I_{SS}}{2nV_T} |V_{id}| \quad (\text{for small } |V_{id}|) \quad (14)$$

However, on semi-log axes ( $\ln |\Delta I_D|$  vs.  $|V_{id}|$ ), in the exponential regime where one transistor dominates conduction:

$$\text{Slope } S = \frac{d \ln |\Delta I_D|}{d |V_{id}|} = \frac{1}{2nV_T} \implies n = \frac{1}{2SV_T} \quad (15)$$

From the measured data points closest to  $V_{id} = 0$  ( $|V_{id}| = 30 \text{ mV}$  to  $60 \text{ mV}$ ):

- For  $V_{id} > 0$ :

$$S_+ = \frac{\ln(7.667 \text{ mA}) - \ln(4.778 \text{ mA})}{0.060 \text{ V} - 0.030 \text{ V}} = \frac{2.0369 - 1.5640}{0.030} = 15.76 \text{ V}^{-1} \quad (16)$$

$$n_+ = \frac{1}{2 \times 15.76 \times 0.02585} = 1.227 \quad (17)$$

- For  $V_{id} < 0$ :

$$S_- = \frac{\ln(11.444 \text{ mA}) - \ln(7.167 \text{ mA})}{0.060 \text{ V} - 0.030 \text{ V}} = \frac{2.4375 - 1.9695}{0.030} = 15.57 \text{ V}^{-1} \quad (18)$$

$$n_- = \frac{1}{2 \times 15.57 \times 0.02585} = 1.242 \quad (19)$$

- Average Measured Slope Factor:

$$n_{\text{meas}} = \frac{n_+ + n_-}{2} = \frac{1.227 + 1.242}{2} = \mathbf{1.235} \quad (20)$$

This extracted value  $n_{\text{meas}} = 1.235$  lies comfortably within the expected physical range of  $1.2 \leq n \leq 2.0$  for subthreshold silicon MOSFETs.

### 8.2 Determination of Transconductance and Voltage Gain

1. Planning Estimate ( $n = 1.5$ , Section 10):

$$g_{m,\text{plan}} = \frac{I_{SS}}{nV_T} = \frac{80.1 \text{ mA}}{1.5 \times 25.85 \text{ mV}} = 2065.8 \text{ mA/V} = 2.066 \text{ A/V} \quad (21)$$

With the actual drain load used in the lab ( $R_D = 180 \Omega$ ):

$$A_{d,\text{plan}} = g_{m,\text{plan}} R_D = 2.066 \times 180 = \mathbf{371.84 \text{ V/V}} \quad (51.4 \text{ dB}) \quad (22)$$

2. Theoretical Prediction with Measured  $n$  ( $n_{\text{meas}} = 1.235$ ):

$$g_{m,\text{thy}} = \frac{I_{SS}}{n_{\text{meas}} V_T} = \frac{80.1 \text{ mA}}{1.235 \times 25.85 \text{ mV}} = \mathbf{2509.0 \text{ mA/V}} = 2.509 \text{ A/V} \quad (23)$$

$$A_{d,\text{thy}} = g_{m,\text{thy}} R_D = 2.509 \times 180 = \mathbf{451.62 \text{ V/V}} \quad (53.1 \text{ dB}) \quad (24)$$

3. Direct Experimental Extraction from Linear  $\Delta I_D$ - $V_{id}$  Plot: Using the two measured data points straddling  $V_{id} = 0$  ( $V_{id} = -30 \text{ mV}$  and  $+30 \text{ mV}$ ):

$$g_{m,\text{meas}} = \left. \frac{\Delta I_D}{\Delta V_{id}} \right|_{V_{id}=0} = \frac{4.778 \text{ mA} - (-7.167 \text{ mA})}{30 \text{ mV} - (-30 \text{ mV})} = \frac{11.944 \text{ mA}}{60 \text{ mV}} = \mathbf{199.1 \text{ mA/V}} = 0.1991 \text{ A/V} \quad (25)$$

$$A_{d,\text{meas}} = g_{m,\text{meas}} R_D = 0.1991 \times 180 = \mathbf{35.83 \text{ V/V}} \quad (31.1 \text{ dB}) \quad (26)$$

### 8.3 Comparison Table and Percentage Deviation

Table 6: Comparison of Small-Signal Transconductance and Voltage Gain (Section 11)

| Quantity              | Planning Estimate ( $n = 1.5$ ) | Theory with $n_{\text{meas}} = 1.235$                                                            | From Measured Slope |
|-----------------------|---------------------------------|--------------------------------------------------------------------------------------------------|---------------------|
| $g_m$ (mA/V)          | 2065.8                          | 2509.0                                                                                           | 199.1               |
| $A_d = g_m R_D$ (V/V) | 371.84                          | 451.62                                                                                           | 35.83               |
| Percentage Deviation: |                                 | $\frac{ g_{m,\text{thy}} - g_{m,\text{meas}} }{g_{m,\text{thy}}} \times 100\% = \mathbf{92.1\%}$ |                     |

### 8.4 Tail Resistance, Common-Mode Gain, and CMRR

- **Common-Mode Gain ( $A_{cm}$ ):**

$$A_{cm} \approx -\frac{R_D}{2R_{SS}} = -\frac{180\ \Omega}{2 \times 22\ \Omega} = -4.091\ \text{V/V} \quad (27)$$

- **Common-Mode Rejection Ratio (CMRR):**

$$\text{CMRR} = \left| \frac{A_{d,\text{meas}}}{A_{cm}} \right| = \frac{35.83}{4.091} = \mathbf{8.76} \implies \text{CMRR}_{\text{dB}} = 20 \log_{10}(8.76) = \mathbf{18.85\ \text{dB}} \quad (28)$$

## 9. Observations and Inferences

1. **Operating Point and Threshold Offset:** At the quiescent bias  $V_{cm} = 3.08\ \text{V}$ , the tail voltage sits at  $V_{SS} = I_{SS} R_{SS} = 80.1\ \text{mA} \times 22\ \Omega = 1.762\ \text{V}$ , yielding an effective gate-to-source bias  $V_{GS} = 3.08 - 1.76 = 1.32\ \text{V}$ . At this bias, the transistors are partially into the moderate inversion regime. The minor difference between  $V_{D1} = 4.55\ \text{V}$  and  $V_{D2} = 4.66\ \text{V}$  (110 mV offset) directly reflects the threshold voltage and transconductance mismatch seen in the individual device characterization curves (Section 4).
2. **Differential Current Asymmetry:** The differential current saturates at +8.61 mA for positive  $V_{id}$ , whereas it reaches -24.44 mA for negative  $V_{id}$ . This asymmetry arises because  $V_{in-}$  was held stationary at 3.08 V while  $V_{in+} = V_1$  was swept. Sweeping  $V_1$  modulates the common-source tail node  $V_S$  through the low tail resistance ( $R_{SS} = 22\ \Omega$ ), which alters the total tail current  $I_{SS}$  dynamically across the sweep rather than maintaining an ideal constant current.
3. **Physical Origin of  $g_m$  Deviation:** While the ideal weak-inversion formula assumes an infinite tail impedance and an ideal current source ( $R_{SS} \rightarrow \infty$ ), the breadboard circuit utilized a passive tail resistor  $R_{SS} = 22\ \Omega$ . Any differential or common-mode swing experiences substantial local source degeneration ( $1 + g_m R_{SS}$  effect) and parasitic lead resistance, which significantly reduces the effective bench transconductance below the idealized un-degenerated exponential formula.
4. **Common-Mode Rejection Behavior:** The measured common-mode gain ( $|A_{cm}| \approx 3.82\ \text{V/V}$ ) closely tracks the theoretical expectation of 4.09 V/V. Because a small resistor ( $22\ \Omega$ ) is used instead of an active current mirror, the CMRR is modest (18.85 dB). In high-performance analog IC design, an active current source with an output impedance of several hundred kilo-ohms is employed, boosting CMRR well past 60 dB.

## 10. Safety Precautions Adhered To

1. **Electrostatic Discharge (ESD):** The SI9926BDY TrenchFET is highly sensitive to ESD due to its ultra-thin gate oxide. The breakout board was handled strictly by its edges.
2. **Current Limiting:** The DC bench power supply was set to a rigid current limit of 150 mA prior to powering the circuit, safeguarding the MOSFETs and breadboard wiring against accidental short circuits.

3. **Maximum Voltage Ratings:** At no point during sweeps did  $V_{GS}$  exceed  $\pm 12$  V or  $V_{DS}$  exceed 20 V, well within the manufacturer's absolute maximum ratings.
4. **Non-Invasive Measurement:** Branch and tail currents were determined exclusively by differential voltmeter measurements across known precision resistors ( $R_D$  and  $R_{SS}$ ), preventing burden voltage artifacts introduced by inline ammeters.

## 11. Conclusion

The DC bias, common-mode transfer characteristic, and differential-mode transfer behavior of a source-coupled MOSFET differential amplifier utilizing the SI9926BDY dual package were experimentally investigated and compared with LTspice simulations. The quiescent point was successfully established at  $V_{cm} = 3.08$  V, delivering  $I_{SS} = 80.1$  mA with balanced drain voltages near 4.6 V. The Input Common-Mode Range was determined to extend from 0.8 V to 3.0 V, bounded by cutoff and triode onset. The measured common-mode gain  $A_{cm} \approx -3.82$  V/V agreed within 6.6% of the theoretical value ( $-\frac{R_D}{2R_{SS}} = -4.09$  V/V). The exponential weak-inversion conduction model was validated on semi-log axes, yielding an extracted slope factor  $n_{meas} = 1.235$ , confirming that carrier diffusion governs conduction at bench currents for this high-density power device.